NewsRadar: Mid-Term Progress Report

Branch: RAHUL · **Repository:** github.com/debord555/cs699_project

1 Introduction

NewsRadar shows Indian news on a map: each story appears as a dot at the place it is about. Phase 1 of the data collection module built the database that every other part of the system reads from.

Part	Role
Data collection (this report)	Gathers articles from Indian news websites and stores them
AI step	Reads each article and finds its location and category
Map website	Displays the articles on an interactive map

Because the AI step and the map both depend on this data, the first phase focused on designing a database that is clean, well-linked and ready to grow.

2 Technology choices

Choice	What it is	Why it was chosen
PostgreSQL	A free, widely used relational database	Handles millions of rows, enforces rules that keep data clean, and supports lists and time zones. Its **PostGIS** add-on understands map coordinates, which the map team will need.
Docker	A tool that runs software inside a sealed “container”	Every teammate gets exactly the same database with one command, on any computer, with no manual installation.
Python	A programming language	Easy to read, with ready-made libraries for reading news feeds and web pages.

3 Database design

3.1 The three tables

The data is split into three linked tables, much like three sheets in a spreadsheet.

Table	Stores	One row means	Example
sources	News websites	One website	The Hindu
feeds	Each website’s list of latest headlines (an RSS feed)	One feed	The Hindu, Kerala news
articles	The news stories	One article	“Floods hit three districts in Assam”

Why three tables instead of one? Details about a website (its name, its region) are stored once in `sources`, not repeated in thousands of article rows. This avoids duplication, keeps the data consistent, and makes changes easy: renaming a website changes one row, not 10,000. This is called normalisation.

3.2 How the tables connect

- One website can have **many feeds**. For example, The Hindu has separate feeds for Kerala, Tamil Nadu, Delhi and others.
- One website publishes **many articles**.
- One feed brings in **many articles**, and every article remembers which feed found it.

These links are made with **keys**:

- A **primary key** (`id`) gives every row a unique number.
- A **foreign key** (such as `articles.source_id`) stores the `id` of a row in another table. The database refuses any link to a row that does not exist, so an article can never point to a missing website.

3.3 Key design decisions

Decision	Reason
A unique fingerprint for each link (`url_hash`)	The same article often arrives through slightly different links. Each link is cleaned and its fingerprint is stored as unique, so the database itself refuses duplicates.
A `status` column	Articles move through clear stages: `new` (found), then `ready` (full text saved), then `llm_done` (processed by the AI step). This is how the module hands work to the AI step without either side getting in the other's way.
Hint columns	Many Indian news feeds cover one region, and Indian stories usually begin with a town name (for example "GUWAHATI:"). Storing these clues helps the AI step place each article faster and more accurately.
Times stored in world standard time (UTC)	Times compare correctly whatever time zone a website uses, which the map's date filters depend on.
Indexes on `status`, `published_at` and `content_hash`	Like the index of a book, they keep the most common searches fast even with millions of articles.
Only links to images are stored	Saves space and avoids copying content that belongs to the news websites.
Full article text kept for internal use only	The AI step needs the full text to understand the story; the map shows only the title, a short snippet and a link to the original site.

4 Relational schema

4.1 sources

Attribute	Type	Key / rule	Description
`id`	SERIAL	PK	Unique number for each website
`name`	TEXT	NOT NULL	Display name, e.g. The Hindu
`domain`	TEXT	UNIQUE, NOT NULL	Web address, e.g. thehindu.com
`language`	TEXT	NOT NULL, default `en`	Main language of the site
`region_scope`	TEXT		Area covered, e.g. national, Assam
`is_active`	BOOLEAN	NOT NULL, default true	Switch to pause a website

4.2 feeds

Attribute	Type	Key / rule	Description
`id`	SERIAL	PK	Unique number for each feed
`source_id`	INT	FK to `sources.id`, NOT NULL	The website this feed belongs to
`url`	TEXT	UNIQUE, NOT NULL	Address of the feed
`kind`	TEXT	NOT NULL, default `rss`	`rss` or `google_news`

Attribute	Type	Key / rule	Description
location_hint	TEXT		Place the feed covers, e.g. Kerala
category_hint	TEXT		Topic the feed covers, e.g. Sports
poll_minutes	INT	NOT NULL, default 30	How often to check the feed
etag	TEXT		Version tag sent by the website
last_modified	TEXT		Time the website says the feed changed
last_polled_at	TIMESTAMPTZ		When the feed was last checked
consecutive_failures	INT	NOT NULL, default 0	Failed checks in a row
is_active	BOOLEAN	NOT NULL, default true	Turns off after 10 failures in a row

4.3 articles

Attribute	Type	Key / rule	Description
id	BIGSERIAL	PK	Unique number for each article
url	TEXT	NOT NULL	Cleaned link to the article
url_hash	CHAR(64)	UNIQUE, NOT NULL	Fingerprint of the link; blocks duplicates
source_id	INT	FK to sources.id, NOT NULL	Website that published it
feed_id	INT	FK to feeds.id	Feed that found it
title	TEXT	NOT NULL	Headline
summary	TEXT		Short description from the feed
body	TEXT		Full text, for the AI step only
dateline	TEXT		Town the story starts with, e.g. GUWAHATI
author	TEXT		Writer
image_url	TEXT		Link to the main picture
language	TEXT		Detected language, e.g. en
published_at	TIMESTAMPTZ		Publication time (UTC)
scraped_at	TIMESTAMPTZ	NOT NULL, default now	When it was collected
source_tags	TEXT[]		The website’s own tags
location_hint	TEXT		Copied from the feed
category_hint	TEXT		Copied from the feed
content_hash	CHAR(64)	Indexed	Fingerprint of the text; finds copies across sites
is_wire	BOOLEAN	NOT NULL, default false	True for news agency copies (PTI, IANS)
status	TEXT	NOT NULL, default new, indexed	Stage: new, ready, llm_done

Attribute	Type	Key / rule	Description
`fetch_attempts`	INT	NOT NULL, default 0	Failed download attempts
`error`	TEXT		Last download error

4.4 Relationships

Relationship	Type	Foreign key	Rule
sources to feeds	One to many	`feeds.source_id`	Every feed belongs to exactly one website
sources to articles	One to many	`articles.source_id`	Every article has exactly one publisher
feeds to articles	One to many	`articles.feed_id`	An article was found by at most one feed

5 Description of the files

File	Purpose
`docker-compose.yml`	Describes the database container: which software to run, where to keep the data permanently, and a health check that confirms the database is ready. Starts everything with one command.
`db/init/01_schema.sql`	Creates the three tables, their links and indexes automatically the first time the database starts.
`.env.example`	A template listing the settings each teammate needs (database name, user, password). Real passwords go in a private `.env` file that is never shared.
`.gitignore`	Tells Git never to upload private or unnecessary files: passwords, installed libraries, logs and data files.
`scraper/requirements.txt`	The Python libraries the project needs, with exact versions, so every computer installs the same thing.
`scraper/db.py`	One shared function that opens a connection to the database using the private settings. All later code uses it.
`scraper/settings.py`	The rules the collector follows: an honest bot name, a 20-second time limit per website, a 3-second pause between visits to the same site, and limits on retries.
`scraper/normalize.py`	Helper tools: cleans links into one standard form, creates link fingerprints, removes formatting from text, and skips pages that are not news stories (videos, photo galleries, live blogs). Includes built-in self-checks.

6 Outcome of Phase 1

- A complete, documented database design for the whole project.
- The database runs in Docker and starts with a single command on any computer.
- Tables, links and indexes are created automatically.
- Shared code for database access and link cleaning is written and tested.
- All work is on GitHub (branch RAHUL).

7 Phase 2 scope

The next phase will involve Google News searches for every Indian state and union territory, compiling a list of news websites and their headline feeds, reading the feeds and saving new articles, finding clues in each article such as the town, news agency and language, and politely downloading the full text of each article. The collector will also be run automatically on a timer, with a health report, and run in Docker.